

Guillaume Derval

PHD IN COMPUTER SCIENCE AND ENGINEERING · POSTDOCTORAL RESEARCHER AT ULIÈGE, BELGIUM

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Research interests

Modeling languages for Combinatorial Optimization (Mixed-Integer) Linear Programming (LP/MILP) Constraint Programming (CP)

- Parametric & robust optimization & sensitivity analysis
- Preprocessing & symbolic modeling
- Parallelization & distribution
- Interaction between solving technologies (CP, MILP, SAT, CFNs)

Explainable AI

- Human-model interaction
- Mathematical model transparency

Software engineering

- Modeling language compilers and interpreters

Energy systems

- Large energy system models

Computer science and mathematics education

- Automated feedback and grading
- Competitive programming

Education

PhD in Computer Science and Engineering [Teaching/Research assistant]

2016-2021 UCLouvain

Thesis: "Finding Maximum Submatrices"

Supervisor: Pr. Pierre Schaus

Master in Computer Science and Engineering [AI & Data science/Applied mathematics]

2014-2016 UCLouvain

Master's thesis: "Parallelization of constraint programming using embarrassingly parallel search"

Magna cum laude

Bachelor in Engineering

2011-2014 UCLouvain

Professional experience

SCK-CEN SMR Chair holder [Postdoctoral researcher]

ULiège

OPTIMIZING THE INTEGRATION OF SMALL MODULAR (NUCLEAR) REACTORS (SMRs) IN THE BELGIAN ENERGY LANDSCAPE

2024-now

- Development of tools and models to assess the possible integration of SMRs in Belgium.
- **Focus on the extension of the GBOML (Graph-Based Optimization Modeling Language) language and the Ora platform to manage uncertainty and robustness considerations.**
- Lead researcher on the project for ULiège.
- Co-supervision of a team of 4 PhD students.
- Grant writing (Horizon Europe, InterReg, PDR, ETF, ...)

Invited Researcher

CRIL, University of Artois, FR

HEURISTICS & MODELS IN CP / APERIODIC TILINGS

Jan-Mar 2025

- At the CRIL laboratory of Pr. Christophe Lecoutre
- Work on CP modeling of aperiodic tiling & heuristics; resulting paper accepted for an oral presentation at AAAI 2026.

Postdoctoral researcher

ULiège

INTEGCER & INTEGRATION PROJECTS (FUNDED BY THE WALLOON REGION / THE ENERGY TRANSITION FUND, BE)

2022-2024

- IntegCER was an industrial project in collaboration with ENGIE, RESA and Haulogy, on the integration of Renewable Energy Communities in Wallonia. The aim of the project is to create the necessary tools and models for the planning of investments and the operational management of such a community, both on an economic level (billing) and a technical level (operations of production and storage means).
- INTEGRATION was an Energy Transition Fund-funded project which led to the creation of the INTEGRATION model of the Belgian Energy System in 2050.
- **Development of GBOML (Graph-Based Optimization Modeling Language).**
- Co-supervision of a team of 4 PhD students.
- Grant writing (Horizon Europe, InterReg, PDR, ...)

Invited lecturer

UCLouvain

LINFO1103 - INTRODUCTION TO ALGORITHMS

2022-2023 and 2023-2024

- Course from the Louvain School of Engineering (EPL), mainly intended for first-year students in computer science.
- Introduction to basic concepts of algorithms and data structures, complexity.

- The course is attended by students from various backgrounds, mostly future computer science bachelors, but also future bachelors in mathematics, biology, architecture, business engineering, and future masters in linguistics. 293 students took the course in 2022-2023, 255 in 2023-2024.

Invited lecturer

LINFO1121 - ALGORITHMS AND DATA STRUCTURES

UCLouvain

2020-2021

- Course from the Louvain School of Engineering, mainly intended for third-year students of civil engineering and computer science bachelor's degrees.
- The course was attended by 224 students.

Research Assistant

THESIS 'FINDING MAXIMUM SUBMATRICES'

UCLouvain

2016-2021

- Finding submatrices with maximum sum; a problem related to biomedical research (DNA micro-arrays).
- Development of new Constraint Programming methods.

Teaching experience

Lecturer	2022-2024	LINFO1103	Introduction to algorithms (FR)	<i>EPL, UCLouvain</i>
	2020-2021	LINFO1121	Data structures and algorithms (FR)	<i>EPL, UCLouvain</i>
Teaching assistant	2023-2024	ELEC0080-1	Energy networks (EN)	<i>FSA, ULiège</i>
	2021-2022	INFO0902	Data structures and algorithms (FR)	<i>FSA, ULiège</i>
	2016-2020	LINFO1121	Data structures and algorithms (FR)	<i>EPL, UCLouvain</i>
	2017-2020	LINFO2365	Constraint Programming (EN)	<i>EPL, UCLouvain</i>
	2019-2020	LEPL1402	Computer Science 2 (FR)	<i>EPL, UCLouvain</i>
	2016-2019	LINFO2132	Languages and translators (EN)	<i>EPL, UCLouvain</i>
	2017-2018	LINFO1113	Numerical algorithms (FR)	<i>EPL, UCLouvain</i>

Projects/grants awarded

2023	FNRS PDR on sensitivity analysis	<i>1 PhD student / 4 years</i>
2024	EPSx tender from the FPS Economy in Belgium, to develop the Ora software	<i>3 FTEs / 1 year</i>
2025	PiSTEM , funded by the Energy Transition Fund	<i>2 PhD students + 0.5 FTE / 3 years</i>

Awards

2017	Distinguished Student Paper Award , for the article "Improved filtering for the bin-packing with cardinality constraint"	<i>CP Conference 2017</i>
2016	ORBEL-Wolsey Award 2016 , Best open-source contribution of the year in Operations Research, for Oscala-Modeling	<i>ORBEL 2016</i>

Publications

I strongly emphasize the repeatability of experiments and results. **My scientific output (software, results, ...) is free and open-source**, and the necessary files for replicating the experiments are available on GitHub or Zenodo.

Published (Peer-reviewed journals, conferences and workshops)

Aperiodic Tiling and Rhythmic Canons: A CP Journey

Christophe Lecoutre, Guillaume Derval

Proceedings of the AAAI Conference on Artificial Intelligence, 2026

Lead-cooled fast reactor SMR integration: An off-grid study case based on a real-life demand

Antoine Larbanois, Bardhyl Miftari, Antoine Mouchamps, Ayyildiz Kerem Enes, Vincent Schryvers, Guillaume Derval, Damien Ernst

Energy Conversion and Management: X, 2026

Towards the integration of small modular reactors in industrial hubs: A multi-dimensional discussion

Antoine Larbanois, Bardhyl Miftari, Guillaume Derval, Damien Ernst

Energy Research & Social Science, 2026

A unified definition of hosting capacity, applications and review

Amina Benzerga, Alireza Bahmanyar, Guillaume Derval, Damien Ernst

IEEE Access, 2025

Integration of offshore energy into national energy system: a case study on Belgium

Jocelyn Mbenoun, Amina Benzerga, Bardhyl Miftari, Ghislain Detienne, Thierry Deschuyteneer, Juan Vazquez, Guillaume Derval, Damien Ernst

Applied Energy, 2025

GBOML: a structure-exploiting optimization modelling language in Python

Bardhyl Miftari, Mathias Berger, Guillaume Derval, Quentin Louveaux, Damien Ernst
Optimization Methods and Software, 2024

Optimal control of renewable energy communities with controllable assets

Samy Aittahar, Miguel Manuel de Villena, Guillaume Derval, Michael Castronovo, Ioannis Boukas, Quentin Gemine, Damien Ernst
Frontiers in Energy Research, 2023

A review of the Constraint Programming MOOC on EdX

Augustin Delecluse, Guillaume Derval, Laurent Michel, Pierre Schaus, Pascal Van Hentenryck
Workshop on Teaching Constraint Programming, WTCP 2023, 2023

Symbolism for modelling, reformulations, and parallelism: MaxiCP-Modelling

Guillaume Derval, Damien Ernst
The 22nd workshop on Constraint Modelling and Reformulation, ModRef 2023, 2023

Maximal-Sum submatrix search using a hybrid constraint programming/linear programming approach

Guillaume Derval, Pierre Schaus
European Journal of Operational Research, 2022

Optimal Decoding of Hidden Markov Models With Consistency Constraints

Alexandre Dubray, Guillaume Derval, Siegfried Nijssen, Pierre Schaus
International Conference on Discovery Science, 2022

Mining constrained regions of interest: An optimization approach

Alexandre Dubray, Guillaume Derval, Siegfried Nijssen, Pierre Schaus
International Conference on Discovery Science, 2020

Nowcasting COVID-19 hospitalizations using Google Trends and LSTM

Guillaume Derval, Vincent François-Lavet, Pierre Schaus
AI 4 Social Good workshop 2020, 2020

The maximum weighted submatrix coverage problem: A CP approach

Guillaume Derval, Vincent Branders, Pierre Dupont, Pierre Schaus
International conference on integration of constraint programming, artificial intelligence, and operations research (CPAIOR 2019), 2019

An aggregate learning approach for interpretable semi-supervised population prediction and disaggregation using ancillary data

Guillaume Derval, Frédéric Docquier, Pierre Schaus
Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML-PKDD 2019), 2019

Mining a maximum weighted set of disjoint submatrices

Vincent Branders, Guillaume Derval, Pierre Schaus, Pierre Dupont
International Conference on Discovery Science, 2019

Improved filtering for the bin-packing with cardinality constraint

Guillaume Derval, Jean-Charles Régim, Pierre Schaus
Constraints Journal (presented at the CP conference 2017), 2018

Embarassingly Parallel Search Reengineered

Guillaume Derval, Pierre Schaus, Jean-Charles Régim
Doctoral Program of the 22nd International Conference on Principles and Practice of Constraint Programming (CP 2016), 2016

Automatic grading of programming exercises in a MOOC using the INGINIOUS platform

Guillaume Derval, Anthony Gego, Pierre Reinbold, Benjamin Frantzen, Peter Van Roy
European Stakeholder Summit on experiences and best practices in and around MOOCs (EMOOCs'15), 2015

Under review (Peer-reviewed journals, conferences and workshops)

Analyzing changes in optimal variables in linear programming with uncertain parameters

Baptiste Istace, Guillaume Derval, Bardhyl Miftari, Quentin Louveaux
arXiv preprint, 2026

ICLF: An Immersive Code Learning Framework based on Git for Teaching and Evaluating Student Programming Projects

Pierre Schaus, Guillaume Derval, Augustin Delecluse
arXiv preprint, 2026

MaxiCP: A Not So Mini Constraint Programming Solver

Pierre Schaus, Guillaume Derval, Augustin Delecluse, Laurent Michel, Pascal Van Hentenryck
Technical report, 2026

Post-Solve Auditing of Solver-Successful Optimization Models with Evidence-Grounded Language Models

Ahmed Tayachi, Bardhyl Miftari, Quentin Louveaux, Damien Ernst, Guillaume Derval
AAAI Conference on Artificial Intelligence 2027, 2026

Sensitivity analysis for linear changes of the constraint matrix of a (mixed-integer) linear program

Bardhyl Miftari, Quentin Louveaux, Damien Ernst, Guillaume Derval
Open Journal of Mathematical Optimization, 2025

Efficient LP warmstarting for linear modifications of the constraint matrix

Guillaume Derval, Bardhyl Miftari, Damien Ernst, Quentin Louveaux

4OR, 2025

Optimal Control of Renewable Energy Communities subject to Network Peak Fees with Model Predictive Control and Reinforcement Learning Algorithms

Samy Aittahar, Adrien Bolland, Guillaume Derval, Damien Ernst

IEEE Access, 2025

Distributed e-Fuel Hubs (DEFH): A Case Study of a Belgian Fischer-Tropsh liquids Hub

Samy Mokaddem, Bardhyl Miftari, Guillaume Derval, Damien Ernst

Applied Energy, 2025

Soon to be submitted (Peer-reviewed journals, conferences and workshops)

Distributed e-Fuel Hubs (DEFH): A Case Study of a Belgian Fischer-Tropsh liquids Hub

Samy Mokaddem, Bardhyl Miftari, Guillaume Derval, Damien Ernst

TBD, 2025

SMR: a numerical model of a Belgian Lead-cooled Fast Reactor (LFR) designed for cogeneration in industrial applications

Antoine Mouchamps, Guillaume Derval, Damien Ernst

TBD, 2025

Presentations without proceedings (at scientific conferences and workshops)

Sensitivity analysis for linear changes of the constraint matrix of an LP/MIP

Guillaume Derval, Bardhyl Miftari, Damien Ernst

Annual Conference of the Belgian Operational Research Society (ORBEL 40), 2026

Stochastic batch scheduling in an industrial context

Florent Hervers, Guillaume Derval, Quentin Louveaux, Xavier Renard

Conference of the International Federation of Operational Research Societies (IFORS 2026), 2026

LLM-Driven Auditing of Optimization Models

Ahmed Tayachi, Quentin Louveaux, Guillaume Derval, Bardhyl Miftari, Damien Ernst

Conference of the International Federation of Operational Research Societies (IFORS 2026), 2026

Analyzing changes in optimal variables in linear programming with uncertain parameters

Baptiste Istace, Guillaume Derval, Bardhyl Miftari, Quentin Louveaux

Conference of the International Federation of Operational Research Societies (IFORS 2026), 2026

Techno-Economic and Life Cycle Assessment of Renewable Hydrogen Production: A Belgian Case Study

Justin Barras, Bardhyl Miftari, Guillaume Derval, Damien Ernst, Angélique Léonard

Conference of the International Federation of Operational Research Societies (IFORS 2026), 2026

Sensitivity analysis for linear changes of the constraint matrix of a (mixed-integer) linear program

Quentin Louveaux, Guillaume Derval, Damien Ernst, Bardhyl Miftari

Conference of the International Federation of Operational Research Societies (IFORS 2026), 2026

The journey of creating a modelling language

Bardhyl Miftari, Guillaume Derval, Quentin Louveaux, Damien Ernst

18th INFORMS Computing Society (ICS) Conference, 2025, 2025

Distributed e-fuel hubs: Concept and case study

Samy Mokaddem, Bardhyl Miftari, Victor Datchet, Guillaume Derval, Damien Ernst

European Conference on Operational Research (EURO 2025), 2025

Sensitivity analysis for linear changes of the constraint matrix of a linear program

Bardhyl Miftari, Guillaume Derval, Quentin Louveaux, Damien Ernst

European Conference on Operational Research (EURO 2024), 2024

Parametric upper and lower bounds of linear variations of a linear problem's LHS

Bardhyl Miftari, Guillaume Derval, Quentin Louveaux, Damien Ernst

Annual Conference of the Belgian Operational Research Society (ORBEL 38), 2024

Efficient exact recomputation of linear modifications of the constraints matrix in Linear Programming

Guillaume Derval, Bardhyl Miftari, Quentin Louveaux, Damien Ernst

European Conference on Operational Research (EURO 2024), 2024

Tutorial: "An Introduction To Sizing And Operations of Energy Systems with GBOML"

Bardhyl Miftari, Guillaume Derval, Damien Ernst

2nd International workshop on "Open Source Modeling and Simulation of Energy Systems" OSMSES 2023, 2023

Exploiting structure in MILP: a modeler's perspective

Bardhyl Miftari, Mathias Berger, Guillaume Derval, Damien Ernst

SIAM OP23, 2023

GBOML: A modelling tool for structured MILPs

Bardhyl Miftari, Guillaume Derval, Mathias Berger, Damien Ernst

Conference of the International Federation of Operational Research Societies (IFORS 2023), 2023

Presentations outside the academic context

Energie: L'avenir des infrastructures gazières et électriques

Damien Ernst, Victor Dachet, Guillaume Derval, Thibaut Téchy

Meeting of the boards, 2025

Energie: De toekomst van de infrastructuur voor gas en elektriciteit

Damien Ernst, Victor Dachet, Guillaume Derval, Thibaut Téchy

Meeting of the boards, 2025

Les communautés d'énergie renouvelables et le partage d'énergie

Guillaume Derval

Conférence Liège Créative 22/02/24, 2024

Optimal Sizing and Operations Of Energy Systems Using GBOML (Course)

Bardhyl Miftari, Guillaume Derval, Damien Ernst

One-day course at ULiège, 2023

Sizing and Operations of Energy Systems Using GBOML

Bardhyl Miftari, Guillaume Derval, Damien Ernst

Company meetings @ULiège, 2023

Hydrogen as the basis of Remote Renewable Energy Hubs

Victor Dachet, Bardhyl Miftari, Guillaume Derval, Damien Ernst

Club Industrie-Université 22/09/23, 2023

Le partage d'énergie en copropriété

Guillaume Derval

Université d'été Climactes 2023, 2023

Contributions to open source scientific tools

GBOML: A Structure-Exploiting Optimization Modelling Language in Python

Bardhyl Miftari, Mathias Berger, Guillaume Derval, Damien Ernst

MaxiCP, a Constraint Programming solver in Java

MaxiCP Team

OscaR: Scala in OR

OscaR Team

Contributions to open source teaching tools

MiniCP

MiniCP Team

JavaGrading

Guillaume Derval, Augustin Delecluse

INGInious

Guillaume Derval, Anthony Géo, Pierre Reinbold, Other contributors

Community work

Program Committee Memberships:

ECAI	European Conference on Artificial Intelligence	2025
LION	Learning and Intelligent OptimizatiON Conference	2025
ECML-PKDD	European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases	2022, 2023, 2024, 2025
AAAI	Conference on Artificial Intelligence	2021, 2023, 2025
CP	International Conference on Principles and Practice of Constraint Programming	2022, 2023, 2024
CPAIOR	International Conference on the Integration of Constraint Programming, Artificial Intelligence, and Operations Research	2022, 2023, 2024
JFPC	Journées Francophones de la Programmation par Contrainte	2022, 2025

Regular reviewer for the journals:

INFORMS	Journal of Computing
EJOR	European Journal of Operational Research
CP	Constraint Programming Journal

